

Causal Set Theory, the Growing Block, and the Buddhist Debate on the Three Times: What the Earliest Vibhajjavādin Records Say

An answer to an invitation in causal-set quantum gravity — and why the tradition that calls itself Vibhajjavāda is its foil, not its twin

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Abstract

This study reads the Buddhist debate on whether past and future things exist (the "three times") against the growing-block view of time that causal set theory's sequential-growth models are offered to support. Rafael Sorkin (2007) remarked that the "accretive time" of these models seems close to C.D. Broad's and to the "Vibhajjavadin" school's; Fay Dowker (2022; revised 2023) invited a scholar to determine that school's view of time from the earliest records. The records, in Pāli, Sanskrit and Chinese, give no single view under the name. In the northern Abhidharma (*Abhidharmakośabhāṣya* ad V.25), the Vibhajjavādins hold that the present and past action whose fruit is not yet given exist, but not past action that has given its fruit, nor the future. That past thins, so the view resembles causal-set accretion only in keeping some past and denying the future; no position in the debate keeps every born element, as a growing block does. In the Pāli tradition, Vibhajjavāda is the Mahāvihāra's self-designation, and its *Kathāvatthu* refutes four rival views of time, among them, in its own words, the northern view's past half, held, its commentary says, by the Kassapikas (Kāśyapīyas). The Mahāvihāra holds presentism about conditioned phenomena, the view Sorkin offers his growth as an alternative to; nibbāna and concepts stand outside the three times. For physics it offers a contrast: past action conditions the present, and a past object is cognized, without either existing. On a second axis, the beginning, the *Paṭṭhāna*'s proximity condition, as the commentaries draw it, gives a stream of mind no first member; bodies and world-regions begin. Classical sequential growth is past-finite; the one past-infinite growth study read (Bento, Dowker and Zalel 2022) builds a consistent covariant framework but offers no dynamics yet that grows a beginningless past in the physical order of its births. Seven findings are listed with their falsifiers.

Keywords: causal set theory, sequential growth, classical sequential growth, past-infinite causal sets, growing block, presentism, eternalism, philosophy of time, temporal becoming, beginningless time, beginningless saṃsāra, Buddhist philosophy, Buddhist philosophy of time, Abhidhamma, Abhidharma, Kathāvatthu, Paṭṭhāna, proximity condition, Abhidharmakośa, Sarvāstivāda, Kāśyapīya, Vibhajjavāda, Theravāda, Mahāvihāra, Vasumitra, Stcherbatsky, Broad, Sorkin, Dowker, Lean theorem prover, defensive publication.

Terms

The paper reads Pāli, Sanskrit and Chinese texts and places them against physics. An examiner or indexer searching in standard terms should find each concept under the term on the right.

Term used here	Standard term
the three times; the three-times debate	the existence of past, present and future things; the debate among early Buddhist schools on eternalism and presentism
<i>sabbam atthi</i> ; Sarvāstivāda (Pāli Sabbatthivāda)	"all exists": the school holding that past, present and future dharmas exist; a form of eternalism
Vibhajjavāda (Pāli), Vibhajjavāda (Sanskrit); Vibhajjavādin	"those who answer by distinguishing": a Buddhist school name; in the Pāli tradition the Theravāda's self-designation; in the <i>Abhidharmakośabhāṣya</i> , holders of the present plus unripened past action
Mahāvihāra; Tambapaṇṇiya	the Theravāda monastic lineage of Sri Lanka whose texts the Pāli canon preserves
Kassapika, Kassapiya (Pāli); Kāśyapīya (Sanskrit)	an early Buddhist school
Andhaka	early Buddhist schools of the Andhra region, so named by the Pāli commentary
<i>Kathāvatthu</i>	<i>Points of Controversy</i> : the fifth book of the Theravāda Abhidhamma Piṭaka, a book of doctrinal debates
<i>Abhidharmakośabhāṣya</i> (the <i>Kośa</i>)	Vasubandhu's treatise on Sarvāstivāda Abhidharma, with his own commentary
<i>dhmma</i> (Sanskrit <i>dharma</i>)	a momentary mental or material phenomenon; an element of existence
presentism about conditioned dhammas	the view that, of phenomena that arise from conditions, only present ones exist
<i>kālavimutta</i> (time-free)	outside past, present and future: in the Theravāda, nibbāna and concepts
<i>Paṭṭhāna</i> ; <i>paccaya</i>	the seventh book of the Abhidhamma Piṭaka; a typed causal (conditioning) relation
proximity condition (<i>anantara</i>)	each mental state conditions the mental state immediately after it
kamma condition	conditioning of a result by a past volitional act, across moments
<i>kammūpacaya</i>	a persisting accumulation of past action (an opponent's doctrine in the Pāli commentary)
stream of mind; mind-stream (<i>santāna</i>)	the succession of one being's mental states
<i>anamatagga</i> ; <i>saṃsāra</i>	without discoverable beginning; the round of rebirths
world-cycle (<i>kappa</i>); contraction	an aeon of Buddhist cosmology; its phase of destruction
sequential growth; classical sequential growth (CSG)	a stochastic dynamics in which a causal set grows by the birth of one element at a time (Rideout and Sorkin)
post; partial post; partial break	in causal set cosmology, an element related to every other element (a bounce); its local analogues (Dowker and Zalel)
alternating growth; alternating transitive percolation (ATP); Z-covtree	growth dynamics for past-infinite causal sets that alternate births into the past and the future; one such dynamics; a covariant framework for two-way growth (Bento, Dowker and Zalel)
discrete general covariance	label invariance of a causal-set growth dynamics: physical predictions do not depend on the order in which elements are born
growing block; accretion	the view that past and present exist and the future does not, so that the sum of what exists grows (Broad 1923)

Findings disclosed

This paper is a study: it discloses readings of texts, not a mechanism. Its findings are listed here at the width that survived review and the census reported below, each with what would break it. The census's surviving width is a **reading of the *Kathāvatthu*'s time debates against Sorkin's and Dowker's causal-set remarks**; findings 1–5 are parts of that reading and nothing wider. The historical identifications they rely on are prior art and are cited as such (Stcherbatsky 1923, p. 43 n.; Vasumitra; Cousins 2001). Findings 6 and 7 were added in the revision of 2026-09-30, after the census, which did not cover them; they assert no priority, since the derivation is the tradition's own and the physics is its authors' (§4.1–§4.2).

1. **Read against Sorkin's footnote, "Vibhajavadin" has two referents, and they meet in one school.** The *Kośa*'s Vibhajjavādin (present plus unripened past action, no future) and the Mahāvihāra's self-designation are different positions. The *Kathāvatthu* §299 refutes the first view's past half in its own words, under the commentary's name *Kassapikā*; Vasumitra gives the Kāśyapīya the *Kośa*'s thesis on past action (§2.3, §5.1). *Breaks if*: the *Kośa*'s Vibhajjavādin is shown to denote the Mahāvihāra specifically, or the §299 thesis is reliably attributed to a school other than the Kāśyapīya/Kassapika.
2. **No position in the Buddhist three-times debate is a growing block.** The *Kośa*'s and the Kassapika's past exists only until an action's fruit is given, and it retains nothing but unripened action; the resemblance to Sorkin's accretion is confined to keeping some past and denying the future (§4). *Breaks if*: a Buddhist source is found that keeps every arisen dhamma as existing past while denying the future; or Sorkin's growth is read so that born elements cease to exist.
3. **The Mahāvihāra's position, read against Sorkin, is presentism about conditioned dhammas: the very alternative he says block-universe arguments presuppose and his growth is meant to escape.** Nibbāna and concepts are time-free (*kālavimutta*) and fall outside the comparison (§3, §5.3). *Breaks if*: a primary Theravāda source affirms that past or future conditioned dhammas exist, including unripened past action, as distinct from their conditioning force.
4. **The Theravāda's account of how the past acts is a contrast with causal-set growth, and it locates the disagreement at the past relatum.** The ceased act conditions its result across moments; it does not exist when the result arises (§299); and nothing persists between them, since the persisting store of kamma (*kammūpacaya*) is the opponent's doctrine in the commentary (§6). *Breaks if*: a primary Theravāda source treats past kamma as existing when it conditions (for example, by accepting a *kammūpacaya*); or the *Kathāvatthu*'s *atthi* ("exists") is shown to be an efficacy predicate, in which case the contrast with the *Kośa* reduces to a difference of wording (§10).
5. **On Dowker's "perception and epistemology", the *Kathāvatthu* holds that a past object is a genuine object although it does not exist** (IX.6, §559; §6). *Breaks if*: §559 is read so that the past object is a present surrogate (a memory-image existing now) rather than the past itself.
6. **Read against causal-set growth, the Pāli texts supply a second axis, the beginning: a stream of mind has no first member, and so neither has beings' wandering, which a mind-stream carries; bodies and world-regions begin.** SN 15.1 says only *not discerned*. The step to *none* rests on the *Paṭṭhāna*'s proximity condition, which its negative method gives every mental state and withholds from matter. The commentaries draw the step themselves, DN 1 §67–68 records the opposite view in order to reject it, and Lean checks the step from that premise (*kāla/Time. lean*). MN 63's undeclared world is, in the commentaries, beings' aggregates asked about as permanent or annihilated, and a beginningless succession of impermanent states affirms neither (§4.1). *Breaks if*: a primary Theravāda source affirms, as its own view, a mental state arising with no proximity condition (a first

citta of a stream, an uncaused ignorance); or a gap in a stream of mind is found that the tradition does not bridge by that condition.

7. **On that axis, the one past-infinite growth study read offers no dynamics yet in which a past without beginning grows in the physical order of its births; and a world-cycle's contraction is neither a post nor a partial post or break.** Bento, Dowker and Zalel's alternating growth gives up birth order as the physical order; its one covariant model is deterministic; their covariant framework has no dynamics chosen on it; and non-deterministic models encourage them. A contraction in the texts is local, as Dowker and Zalel's partial posts and breaks are, but the re-evolved world is peopled from streams outside the collapse, so its past is not closed (§4.2). *Breaks if:* a growth dynamics for past-infinite causal sets keeps birth order as the physical order, satisfies discrete general covariance and is not deterministic with respect to its observables (their models (18) meet only the last); or every stream in a re-evolved world is read as passing through the contraction.

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Disclosed in the first version (2026-09-27): the two-referent reading of Sorkin's "Vibhajavadin", the placement of the Mahāvihāra's presentism as the foil of causal-set growth, and the contrast of influence without existence. The revision of 2026-09-28 corrected the placement of the Kośa's view (a past that thins, not a growing block; §9) and listed the findings; the revision of 2026-09-30 added §4.1–§4.3 and findings 6 and 7. The date of any passage is that of the earliest timestamped version carrying it.

What is already public, and cited rather than claimed:

- **The two-referent answer in outline.** Stcherbatsky, *The Central Conception of Buddhism* (1923), p. 43 and its footnote, is the nearest prior instance, found after the census below. It describes the Kośa's Vibhajyavādins (citing AK V.24 ff.); it reports that the *Kathāvatthu* ascribes such opinions to the Kāśyapīyas and, citing McGovern, that Xuanzang's commentary identifies the Kośa's Vibhajyavādins as the Kāśyapīyas; it remarks that the Theravādins seem to share the Sautrāntika view; and it notes the double sense of *vibhajyavāda*. This is the page Dowker cites.
- **The Kāśyapīya's thesis in the northern doxography.** Vasumitra's *Samayabhedoparacanacakra* (Xuanzang's Chinese, T49 no. 2031) gives the Kāśyapīya the Kośa's thesis on past action.
- **The four debates in English.** Shwe Zan Aung and C.A.F. Rhys Davids, *Points of Controversy* (Pali Text Society, 1915), translate the *Kathāvatthu* with the commentary's attributions. The Kośa passage is translated by La Vallée Poussin and, in English, by Pruden (1988–90).
- **The mapping of the Buddhist three-times debate onto Western theories of time.** Kristie Miller classes the Theravādins and Sautrāntikas as presentists and the Sarvāstivādins as eternalists, adding that the Sarvāstivāda view is closer to a moving spotlight than to a block universe ("A Taxonomy of Views about Time in Buddhist and

Western Philosophy," *Philosophy East and West* 67.3, 2017). The Vibhajjavāda and Kāśyapiya positions are surveyed in an unpublished essay by Michel Gauvain, *Time and Temporality in Indian Buddhism* (academia.edu 43833572). Its gloss of Vibhajjavāda ("time permanent, phenomena move from one time to another") differs from the Kośa's.

- **The historical claim that Vibhajjavādin named a group of schools, not one** (L.S. Cousins, "On the Vibhajjavādins," *Buddhist Studies Review* 18.2, 2001).
- **The growing block, and causal-set growth as a growing block.** Broad states the view in *Scientific Thought* (1923, pp. 66–67): it "accepts the reality of the present and the past, but holds that the future is simply nothing at all," and "the sum total of existence is always increasing." Tooley (*Time, Tense, and Causation*, 1997) defends a version of it, as Arageorgis classes him. Earman ("Reassessing the Prospects for a Growing Block Model of the Universe," *International Studies in the Philosophy of Science* 22.2, 2008) holds that the causal-set approach gives the growing block new life (as quoted by Arageorgis; not read directly). Arageorgis ("Spacetime as a causal set: Universe as a growing block?", *Belgrade Philosophical Annual* 29, 2016) argues the contrary: that, mainly because of discrete general covariance, the program supports a growing block no better than relativity on a continuum does. This case cuts against the label this paper uses for Sorkin's growth; §4 states what the comparison rests on instead.
- **Causal-set becoming as a philosophical topic** (Sorkin 2007; Wüthrich & Callender, "What Becomes of a Causal Set?", *BJPS* 68.3, 2017; Dowker 2022/2023).
- **Growth dynamics for past-infinite causal sets**, the physics findings 6 and 7 are placed against: Bento, Dowker and Zalel, "If time had no beginning: growth dynamics for past-infinite causal sets" (*Classical and Quantum Gravity* 39, 2022, 045002). The finite-past growth it generalizes is Rideout and Sorkin's (2000); posts and cycles are those of Martín, O'Connor, Rideout and Sorkin (2001) and, as partial posts, breaks and cycles, of Dowker and Zalel (2017, 2023). Bento, Dowker and Zalel credit Wüthrich and Callender (2017) with the suggestion that growth might be compatible with a past-infinite causal set.
- **Beginningless saṃsāra and its derivation from conditionality** are the tradition's own (SN 15; AN 10.61 with *Visuddhimagga* XVII §584; the commentary on MN 9; the *Atthasālinī*; §4.1). This paper reports the derivation; its own parts are the Lean check (CC0, doi:10.5281/zenodo.23067241) and the causal-set placement. *The Wheel That Unwinds the Wheel* (abhidhamma-executable-process-specification), in this corpus, states that the mindstream is beginningless.
- **Buddhist beginninglessness set against physical cosmology**, in popular books: Ricard and Thuan, *The Quantum and the Lotus* (2001); the Dalai Lama, *The Universe in a Single Atom* (2005). Known here from a review and an encyclopedia summary; not read.
- **General comparisons of the Abhidhamma with causal sets or quantum physics**: a blog post of 2026-07-31, self-described as AI-generated, "Causal Set Theory and Abhidhamma: A Dialogue Between Modern Physics and Buddhist Philosophy" (<https://ekasiswanto.wordpress.com/2026/07/31/causal-set-theory-and-abhidhamma-a-dialogue-between-modern-physics-and-buddhist-philosophy/>), and *The Ephemeral Projector: A Comparative Analysis of Quantum Mechanics and Abhidhamma* (ResearchGate publication 397427972, <https://www.researchgate.net/publication/397427972>).

The census. A quick novelty census ran on 2026-09-27. Its pre-registration was pushed publicly before any query: `thonly/publications commit 7348ba2, timestamps/census/physics-lens-prereg.md`. Its aperture was the English web, arXiv, Google Scholar, PhilPapers, GitHub and Google Patents by keyword. Thai, Burmese, Sinhala, Chinese, Japanese and Khmer sources were **NOT RUN**, and books were not read. **Not found in that aperture on that date**: a reading of the *Kathāvatthu*'s time debates against Sorkin's and Dowker's causal-set remarks. The census also reported the two-referent answer as not found; that row is **narrowed** here, because Stcherbatsky's footnote, read

after the census, already gives most of it. A census result is a statement about a search, not a claim of priority. The comparison of Buddhism with physics is a crowded field, and much of it is unindexed. Findings 6 and 7 were added after this census and were not searched; they claim no priority (see *Findings disclosed*). The growing-block literature on causal sets (Earman 2008; Arageorgis 2016) was found at the revision of 2026-10-06 and is cited above; it does not touch the reading the census reports.

Note. As first published on 2026-09-27 this paper stated that its authors and HeartBank® would not seek patent on anything in it ("*The authors and HeartBank® will not seek patent on anything herein*"); it carried no commitment not to assert. The commitment not to assert, and the extension of both commitments to the other entities named above, were added, and the statement brought to the standard form above, on 2026-10-06. The omission did not affect the paper's standing as prior art, which publication establishes; the date relied on is the one carried by this document's OpenTimestamps proof.

1 • The Invitation

Causal set theory replaces the spacetime continuum with a locally finite partial order: discrete events, related only by *before*. Sorkin's slogan is *order + number = geometry*. In its sequential-growth ("classical sequential growth") models the causal set is not given all at once; it **grows**, one element at a time, each new element born to the future of some of those already present. Sorkin argued that this growth gives an objective correlate of the passage of time, a becoming that relativity was thought to have ruled out. His 2007 paper, "Relativity theory does not imply that the future already exists: a counterexample," makes the case. In §5 he describes what the growth yields as "*something 'four dimensional' from the very beginning, but which at any stage of its growth is still incomplete*", and he adds the footnote:

The notion of "accretive time" that arises here seems close to that of C.D. Broad, and also to that of the "Vibhajavadin" school within the Buddhist philosophical tradition.

Fay Dowker, developing Sorkin's proposal as a response to the hard problem of consciousness (arXiv 2209.07653), quotes the footnote and adds her own. In the revision of 21 May 2023 it reads:

A interesting project for a scholar of Buddhism interested in physics would be to try to determine the "Vibhajavadin" view of time, perception and epistemology in detail by examining the earliest existing records.

(The first version, of 16 September 2022, reads "*would be to determine the 'Vibhajavadin' view*", without *try to*.)

For the Vibhajavādin view her notes point to p. 43 of Stcherbatsky's *The Central Conception of Buddhism* (1923). Separately, they cite his *Buddhist Logic* (vol. 1, 1932, pp. 4–5) for early Buddhism as "no Matter, no Substance, only separate elements, momentary flashes of efficient energy ... a flow of existential moments."

The question is well posed, and it is answerable, because the records exist, in Pāli, Sanskrit and Chinese, and they are unusually explicit about time. What follows reads them.

Spelling note. Sorkin and Dowker's 2023 revision write *Vibhajavadin*; Dowker's 2022 version writes *Vibhajyavadin*. They are quoted as written. Pāli sources give *Vibhajjavāda*, Sanskrit sources *Vibhajyavāda*. The spellings mark traditions, and this paper keeps them apart deliberately.

2 · Who Were the Vibhajjavādins?

2.1 · The Pāli self-designation

The *Kathāvatthu* itself, the fifth book of the Abhidhamma Piṭaka, never uses the word. The self-designation is found in its commentary (*Kathāvatthu-aṭṭhakathā*, part of the *Pañcappakaraṇa-aṭṭhakathā* traditionally ascribed to Buddhaghosa, c. 5th century CE, drawing on older material). The commentary describes the Third Council under Aśoka: the king, having expelled monks of other views, asks the remaining monks what the Buddha taught.

"*kiṃvādī, bhante, sammāsambuddho*"ti? "*Vibhajjavādī, mahārājā*"ti.

"Of what doctrine, venerable sir, was the Fully Awakened One?" — "A Vibhajjavādin, great king."
(*abh03a.att.xml:2181, aṭṭhakathā*)

The elder Moggaliputtatissa then expounds the *Kathāvatthu* to refute the views that had arisen and those that would arise, setting out, the commentary says, a matrix the Buddha had laid down for the purpose (:2181 ; :1949). In the tradition's own account, therefore, **the *Kathāvatthu* is the Vibhajjavādin record.**

2.2 · A name for a group

Cousins (2001) shows that the Pāli sub-commentators (the *ṭīkā* writers) know two related senses of the word: an epithet of the Buddha (one who answers by analysis, *vibhajja*) and the name of the Buddha's followers, specifically of their own school. Historically, he argues, it is the name for the viewpoint of the common ancestors of **four** schools: the Mahīṃsāsakas, the Dhammaguttikas (the commentary's form; Cousins's title writes *Dhammaguttaka*), the Kassapiyas and the Tambapaṇṇiyas, the last being the Mahāvihāra of Sri Lanka whose texts the Pāli canon preserves. The commentary's Third Council story, Cousins argues, draws its force from the double meaning. The present-day Theravāda is therefore **one branch** of the Vibhajjavādins, not the whole group. That matters below, because another branch reappears as an opponent.

The commentary's own family tree is different, and it is recorded here because it cuts against a tidy reading. Its schism narrative, which it supports with verses quoted from the *Dīpavaṃsa*, derives the Sabbatthivādins and the Dhammaguttikas from the Mahīṃsāsakas and then says: "*Puna sabbatthivādakulato bhijjivā kassapikā nāma jātā*" ("again, breaking from the Sabbatthivāda family, those called Kassapikas arose"; *abh03a.att.xml:1961*). The Pāli commentary thus files the Kassapikas under the Sabbatthivāda, where Cousins places them among the Vibhajjavādin branches. The findings below depend on *which school* held the §299 thesis, not on which family tree is right.

2.3 · The northern definition [cross-tradition]

The Sarvāstivāda's own systematization, as reported and criticized by Vasubandhu, defines the two camps sharply (*Abhidharmakośabhāṣya* ad V.25, Pradhan ed., p. 296, ll. 4–6):

*ye hi sarvam astīti vadanti atītam anāgataṃ pratyutpannaṃ ca te sarvāstivādāḥ / ye tu kecid asti yat
pratyutpannaṃ adattaphalaṃ cātītaṃ karma kiñcin nāsti yad dattaphalaṃ atītam anāgataṃ ceti vibhajya vadanti
te vibhajjavādinaḥ /*

Those who say that *all exists*, past, future and present, are Sarvāstivādins. Those who, *distinguishing* (*vibhajya*), say that something exists (the present, and past action that has not given its fruit) and something does not (past action that has given its fruit, and the future) are Vibhajjavādins.

(The text is GREIL's, after Pradhan, with word division added; two typographical slips in the e-text, *ye he* and *anāmataṃ*, are given as *ye hi* and *anāgataṃ*.)

The *Kośa*'s Vibhajyavādin, then, holds **present plus some past, and no future**. The name is glossed from the method: they *speak distinguishing*. The verse goes on to count four kinds of Sarvāstivādin, by how a dharma differs as it passes through the three times: in mode of being, in characteristic, in state, or in relation, the views reported (*kila*) of Dharmatrāta, Ghoṣaka, Vasumitra and Buddhadeva. The bhāṣya judges the third, Vasumitra's, the best, under which the times are fixed by a dharma's activity (*kāritra*): future before it acts, present while it acts, past once it has acted; it then presses objections against the Sarvāstivāda position as a whole (V.25d–26 and after, Pradhan pp. 296–297). The Vasumitra named there is a Sarvāstivāda master; this paper does not decide whether he is the doxographer of the same name cited below.

Stcherbatsky's p. 43, the page Dowker cites, describes the same school: an intermediate position that "maintained the reality of the present facts and of that part of the past ones which had not already lost their influence, but the reality of the future ones and of that part of the past ones which had ceased to exhibit any influence it denied. Vasubandhu calls this school the Vibhajyavādins." **Dowker's pointer and the *Kośa* therefore agree**. Stcherbatsky's footnote goes further. It reports that the *Kathāvatthu* (he cites "i, 8") ascribes such opinions to the Kāśyapīyas, who also admitted the reality of that part of the future which was foreshadowed or fixed; that, according to McGovern, Xuanzang's commentary identifies the *Kośa*'s Vibhajyavādins as the Kāśyapīyas; that the Theravādins seem to have shared the Sautrāntika view; and that Vasubandhu seems not to know the explanation of *vibhajyavāda* as orthodoxy, the sense the *Kathāvatthu* commentary gives it.

Vasumitra's doxography of the schools, in Xuanzang's Chinese, gives the Kāśyapīya (飲光部) the same thesis on past action. Among its tenets: 若業果已熟則無。業果未熟則有 ("if the fruit of an action has ripened, it does not exist; if the fruit has not ripened, it exists"). Beside it stands a second criterion of the same shape: a dharma already abandoned and fully known does not exist, and one not yet abandoned and fully known does. The list adds that some formations have a past thing as their cause and none a future thing (無諸行以未來為因), and that all formations perish momentarily (一切行皆剎那滅). It states no thesis on whether future things exist.

The *Kośa*'s definition is a Sarvāstivādin doxographer's report of an opponent (§10). Two northern reports, the *Kośa* and Vasumitra, agree on its past half (both from a Sarvāstivāda milieu, so not fully independent), and the Pāli debate book states the same past-half in its own words (§3).

3 • The Earliest Pāli Record: Four Debates on Time

The *Kathāvatthu* is a book of debates. The Theravādin (*sakavādī*) cross-examines an opponent (*paravādī*) by paired questions until the opponent's thesis yields a consequence he must reject. Four early debates, all in its first chapter, concern time. The *Kathāvatthu* names no opponents; the attributions below are the commentary's. Paragraph numbers follow the Chatṭha Saṅgāyana edition, whose numbering of the debates differs from the one in common scholarly use: CST numbers the four debates 1.5, 1.6, 1.7 and 1.9, while SuttaCentral numbers them I.6, I.7, I.8 and I.10, and Stcherbatsky cites the third as "i, 8". The § numbers are unambiguous. An English translation with the attributions is *Points of Controversy* (1915).

CST §	Debate	Opponent's thesis	Commentary's attribution
282	<i>Sabbamatthitīkathā</i>	<i>sabbam atthi</i> : all exists	" <i>seyyathāpi etarahi sabbamatthivādānaṃ</i> " (the all-exists school, i.e. the Sarvāstivāda)
297	<i>Atītakkhādhādikathā</i>	past and future things are aggregates, bases and elements, and therefore exist	the paravādin who "wants past and future to exist because they do not give up being aggregates"
299–300	<i>Ekaccaṃ atthitīkathā</i>	some of the past exists and some of the future (what is bound to arise). The restriction to action whose result has not ripened is in the <i>Kathāvatthu</i> 's own text: " <i>atītā avipakkavipākā dhammā – te atthi</i> "	" <i>seyyathāpi kassapikā</i> " (the Kassapikas; one of Cousins's four Vibhajjavadin branches, though the commentary's schism list derives them from the Sabbatthivāda, §2.2)
304	<i>Hevatthikathā</i>	things exist <i>in one way</i> and not in another: the past exists <i>as past</i>	" <i>seyyathāpi etarahi ... andhakānaṃ</i> " (the Andhakas)

The refutations of *all exists* and *some exists* turn on one formula:

Atītaṃ atthīti? Āmantā. Nanu atītaṃ niruddhaṃ vigataṃ vipariṇataṃ atthaṅgataṃ abbatthaṅgantanti? (§283)

"Does the past exist?" — "Yes." — "But has the past not ceased, departed, changed, set, gone down?"

The other two debates are refuted differently.

- **§297 separates classification from existence.** The Theravādin grants that past things are *classified* as aggregates but denies that they *exist*. The *Kathāvatthu* itself quotes the *Niruttipatha Sutta* (SN 22.62) in the *Sabbamatthi* and *Atītakkhādhādi* debates (§296, §298): what has passed is designated "*was*", never "*is*" or "*will be*". The opponent's sutta is SN 22.48 (any form, past, future or present, is the form aggregate), cited first at §296 and in reply at §298. The commentary applies the *Niruttipatha* to §297 and says the opponent's sutta proves the aggregates' *being aggregates*, not their existence. That sutta anchors the view in the discourses (the Nikāyas), a layer older than the debate book.
- **The *Hevatthi* thesis grants that the past has ceased and says it exists *as past*** ("*atītaṃ atītanti hevatthi, atītaṃ anāgatanti heva natthi*", §305). The ceased-formula cannot touch it. It is refuted by a different reductio: if the same past both is and is not, then existence and non-existence become one meaning ("*atthaṅṅṅho natthaṅṅṅho ... ekaṅṅṅhe same samabhāge tājāte*", §304–305).

At §299 the Kassapika position is broken apart. If some of the past exists and some does not, then, the commentary explains, some of the past has ceased and some has not, which the opponent will not say; and the opponent grants that past action whose result has not ripened has ceased, so it cannot be said to exist (abh03m3.mul.xml:3345 – :3365; abh03a.att.xml:3217). The same dilemma is run for the future at §300.

The position that remains, the Mahāvihāra's own, is **presentism about conditioned dhammas**: of conditioned phenomena only present ones exist; past and future are what *was* and what *will be*. The Abhidhamma adds a class that the three times do not cover. The root verse of *Abhidhammatthasaṅgaha* ch. 8 §37, summing up the states that serve as the twenty-four conditions, reads "*tekālikā dhammā, kālamuttā ca*", things of the three times and things free of time, and Sumaṅgala's sub-commentary names the time-free class: nibbāna and concepts (*nibbānapaññattivāsena kālavimuttā ca*). Nibbāna is not a present dhamma; it is outside the question this debate asks.

4 • The Physicists' Map and the Buddhist Map

WHAT EXISTS?	past	present	future	nearest physics picture
Sarvāstivāda	all	all	all	eternalism: a moving spotlight, not a block (Miller 2017) †
Kośa's Vibhajjavāda [X]	unripened karma only	all	none	a past that THINS; resembles Sorkin's only in keeping some past and denying the future
Kassapika (Kv §299–300)	unripened karma only	all	what is bound to arise	(no clean analogue: a past that thins, a future that is partly settled)
Andhaka (Kv §304)	"as past"	all	"as future"	a tensed-property reading
Mahāvihāra (Kv, refuting all four)	none	all	none	PRESENTISM about conditioned dhammas ← what Sorkin says his model is NOT ‡
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Sorkin's growth (for comparison)	every born element	no single global "now"	none	growing block (accretion)

No Buddhist row keeps every born element.

† Miller (2017) classes the Sarvāstivādins as eternalists and says explicitly that they are not block-universe theorists; their view looks like a moving spotlight. Cited, not claimed.

‡ Of conditioned dhammas. Nibbāna and concepts are time-free (kālavimutta), outside all rows.

[X] cross-tradition (northern) source.

Sorkin's growth is not presentism. In the sequential-growth models, what has been born **stays**: the causal set at any stage contains its whole past and lacks only its future. Sorkin's own words set the view against a present-only doctrine. Most arguments for the block universe, he writes, "*presuppose that the sole alternative to the 'block universe' is a doctrine that identifies reality with a three-dimensional instant*". He also denies that the process has any single "now" spreading over it. Wüthrich and Callender (2017) examine what kind of becoming this is. They conclude that a novel, *global* becoming is compatible with causal sets, distinct from the *localized* becoming that the theory's advocates consider compatible with its dynamics. Nothing in that literature reads the model as presentist.

Whether the growth supports a growing-block metaphysics is itself disputed. Earman (2008, as quoted by Arageorgis) holds that the causal-set approach gives the growing block new life; Arageorgis (2016) argues that, mainly because of discrete general covariance, the program supports a growing block no better than relativity on a continuum does, and Sorkin himself writes that stopping the growth at a given stage "has no objective meaning within the theory". This paper's placement does not depend on that dispute. It uses only a feature of the models that neither side questions: a born element is never removed. "Growing block" in the map above names that feature, not a verdict on the metaphysics.

On the Buddhist map, then, no row is a growing block. The nearest is the *Kośa's* Vibhajjavādin, and only in two respects: it keeps some past, and it denies the future. Its past thins. An action ceases to exist once its fruit is given, and no other past dhamma is kept at all, where Sorkin's causal set keeps every element ever born. Sorkin's footnote points at a real resemblance in the school the northern sources describe, narrower than the word "close" suggests. It does not describe the school that uses the name today.

4.1 • A second axis: the beginning

The map above has one axis: what exists. The Pāli texts and causal-set growth also differ on a second, which Sorkin's footnote did not raise: **whether the order of what has happened has a first member**. In the texts that order belongs to each stream of mind; there is no global one. This section and the next two were added in revision (2026-09-30).

What the texts say. The *Anamatagga-saṃyutta* (SN 15) opens:

Anamataggoyaṃ, bhikkhave, saṃsāro. Pubbā koṭi na paññāyati avijjānīvaraṇānaṃ sattānaṃ taṇhāsaṃyojanānaṃ sandhāvataṃ saṃsarataṃ. (SN 15.1; CST s0302m.mul.xml:3489, mūla)

"Without discoverable beginning, bhikkhus, is this wandering on. A first point is not discerned of beings hindered by ignorance and fettered by craving, running on and wandering."

The verb is epistemic. The commentary renders it "the prior limit is not seen" (*purimamariyādā na dissati*), glosses *anamatagga* as "with prior and later ends undelimited" (*aparicchinnapubbāparakoṭiko*), and reads the wandering as "the unbroken succession of the aggregates" (*khandhādīnaṃ avicchinappavattā paṭipāṭi*; s0302a.att.xml:2489, aṭṭhakathā). That reading is the commentary's; the *Kathāvatthu* quotes SN 15.1 only as the Puggalavādin's proof-text for a transmigrating person (§75, abh03m3.mul.xml:849). SN 15.5's "not easy to count" concerns the years in one aeon, SN 15.7–15.8's the aeons past (s0302m.mul.xml:3529, :3553, :3565; the sibling paper, §5.3). The commentary ascribes the difficulty to the wandering's length (*dīghatamattā*) only as the questioning monk's reported thought (*kira*, s0302a.att.xml:2521).

What the texts leave undeclared. MN 63 leaves undeclared whether the world is eternal or not, finite or infinite ("sasso loko ... anantavā loko", s0202m.mul.xml:1397, mūla). There, the commentaries say, the world is the world of beings (*sattaloko*; vin01a.att.xml:2169, s0516a.att.xml:881), and the *Atthasālinī* gives the predicates: the five aggregates held "permanent, stable, for all time" (*nicco dhuvo sabbakāliko*) or "cut off, perishing" (*ucchiṇṇati vinassati*), and finite or infinite by the edge of a meditation device (*kaṣiṇaparicchedantena*), in space (abh01a.att.xml:9149). **So the line runs by predicate, not by subject.** A beginningless succession of impermanent states, any stream of which ends at an arahant's death, affirms neither permanence nor annihilation and says nothing of extent (an interpretation; §10). The predicate, not an unseen end, also separates it from DN 1's eternalist, whose self and world stand "like a peak" (*kūṭaṭṭho*, s0101m.mul.xml:283): the DN sub-commentary files even the fortuitous-origination view (below) under eternalism for seeing no end ahead (*anāgate koṭiā dassanato*, s0101t.tik.xml:2061, tīkā).

From "not discerned" to "none", for mind. The *Paṭṭhāna* makes each earlier state a proximity condition of the next (abh03m7.mul.xml:85 – :93, mūla), and its general statement confines the condition to mind:

Yesaṃ yesaṃ dhammānaṃ anantarā ye ye dhammā uppajjanti cittacetasikā dhammā, te te dhammā tesaṃ tesaṃ dhammānaṃ anantarapaccayena paccayo. (abh03m7.mul.xml:97)

"Whatever states, mind and mental factors, arise immediately after whatever states, those states are a condition by proximity for them."

Its negative method lists what arises *not* by proximity, and the rows it gives are all matter: mind-born, kamma-born at rebirth, the great elements, nutriment- and temperature-born, and the mindless beings' (:1201, referring to :1153 – :1169; later chapters abbreviate or refer back). No mental state is listed there, not even a rebirth mind. That is the premise, **a proximity predecessor for every mental state**, which a first member of a stream of mind would lack. Conditionality in general would not suffice: fifteen of the twenty-four conditions are present-only (§6), and a mental state whose conditions were all co-nascent would be conditioned and still first. Matter differs. A body begins at each rebirth, and a world-region's matter begins again after a fire that burns until not an atom's worth of formation remains (*Visuddhimagga* XIII §405–406, e0102n.mul.xml:769 – :773), that contracting world being the receptacle world (*bhājanaloka*, s0101t.tik.xml:1945). **The claim is of streams of mind, and of beings' wandering because a mind-stream carries it; bodies and world-regions begin.**

The step is the tradition's own. The commentary on MN 9 derives it from earlier ignorance conditioning later ignorance: "the beginninglessness of saṃsāra is established" (*saṃsārassa anamataggaṭā sādhitā hoti*, s0201a.att.xml:4309, aṭṭhakathā). The *Visuddhimagga* denies that ignorance is an uncaused root ("Na akāraṇaṃ", XVII §584, e0102n.mul.xml:3553), giving its cause from MN 9 ("with the arising of the taints, the arising of ignorance"), and in the same paragraph cites AN 10.61 (e0102n.mul.xml:3557): "a first point of ignorance is not discerned ... and yet it is discerned: ignorance has this as its condition" (*idappaccayā avijjā*, s0404m3.mul.xml:1890, mūla). The *Atthasālinī* turns *not discerned* into *none*: no boundary exists, after so many hundreds of thousands of aeons or in some king's or buddha's time, before which beings were not ("*ayaṃ paricchedo natthi*", abh01a.att.xml:409). The aeons behind beings therefore exceed every count, not only every easy one. This paper reports the step; it does not claim it.

The view that holds the other option. DN 1 records a stream with a first member, to reject it. A being falls from the mindless deities (*asaññasattā*), recollects the arising of perception and nothing before it, and concludes "*Ahañhi pubbe nāhosim, somhi etarahi ahutvā santatāya pariṇato*", "I was not before; now, not having been, I have come to be" (§67–68, s0101m.mul.xml:483–:487, mūla). The DN sub-commentary answers with the proximity condition: a mind long ceased "is merely a proximity condition, not the seed; the seed is kamma" (*anantarapaccayamattaṃ hotiyeva, na bjaṃ, bjaṃ pana kammaṃ*, s0101t.tik.xml:2057, ṭikā). The same bridge spans emergence from cessation (abh03m7.mul.xml:5673) and death to rebirth across thousands of aeons (*na kāladūratā*, Kv anuṭṭikā, abh05t.nrf.xml:2409). DN 27's re-evolving world starts a world, not a stream: its first beings fall from the Ābhassara heaven (s0103m.mul.xml:1105).

The step, checked. From the premise the step is short, and Lean 4 checks it (`kala/Time.lean`, theorem `beginningless`, in the CC0 repository `SiliconWat/formal-abhidhamma`):

Conditioned s	:=	$\forall m, \exists p, \text{succ } p = \text{some } m$	every moment succeeds a moment of the same stream
tick_succ	:	$\text{succ } m = \text{some } n \rightarrow \text{tick } n = \text{tick } m + 1$	
beginningless	:	$\text{Conditioned } s \rightarrow \neg \exists \text{ first}, \forall m, \text{tick first} \leq \text{tick } m$	

`Conditioned` is the proximity premise for one stream of mind, not conditionality in general, and **the content is the premise**: replace it by `True` and the theorem fails (one of the file's deliberate breaks), while a model with a last moment and no first satisfies every axiom. For a presentist, "no first member" is a claim about what was (§3): a stream's has-beens have no first term; no infinite collection of past states exists now.

4.2 · Where causal-set growth stands on the beginning

Page numbers are those of *Bento*, Dowker and Zalel (*arXiv:2109.10749v1*; published 2022), read in full, unless another paper is named.

Finite-past growth has a beginning "by definition of the model" (p. 3): in classical sequential growth no element is born to the past of an existing one, and "there are no past-infinite causets at all in the sample space" (p. 9). Read a stream's moments as a chain, as the sibling paper's relation *S* does (§3.3 there): a chain with no first member gives each member an infinite past, which no causal set in that sample space has.

Bounces, forward from a beginning. Transitive percolation (TP) almost surely grows infinitely many *posts*, elements related to every other (p. 15), each "a kind of collapse of the universe to zero diameter followed by a subsequent re-expansion" (Martín et al. 2001, p. 3); "TP realises the heuristic of a bouncing universe with a beginning" (p. 15). Collapse need not be global. Dowker and Zalel (2017, pp. 3–4) model a black hole birthing a universe beside an unrelated region by a *partial post* (an element *y* such that if *y* < *x* and *y* is unrelated to *z*, then *x* is unrelated to *z*) or a

partial break (A, B) (every element of A before every element of B , B a future set, $A \cup B$ a past set), and cosmic renormalisation follows. Dowker and Zalel (2023) model cycles by breaks as well as posts. All of this is past-finite.

Growth for past-infinite causal sets. Bento, Dowker and Zalel ask whether growth is *compatible* with a past without beginning; their process still starts "from the empty set" (p. 9), where the texts want no first stage. Four results bear on the texts.

1. **Growth into the past gives up becoming.** Elements must be born to the past of existing ones, "thus making it (nearly if not entirely) impossible to conceive of the growth process as a physical process of Becoming" (pp. 3–4). The alternation of forward and backward births is fixed "for definiteness" (p. 10); random variants (p. 12, n. 8) were not surveyed.
2. **The one covariant alternating model is deterministic.** Among alternating counterparts of the classical models, only alternating transitive percolation (ATP) satisfies discrete general covariance (Claim 3.5, p. 16). It realises "a bouncing universe with no beginning" (p. 15), but every proposed observable (a convex-event) has probability 0 or 1 in it (p. 24): "ATP is not a useful cosmological model" (p. 25).
3. **Other alternating models are not deterministic** (Claim 3.9, p. 25, the couplings of their equation (18)), and "the existence of non-deterministic alternating growth models encourages us to continue" (p. 26). By Claim 3.5 they are not covariant.
4. **A consistent covariant framework exists, with no dynamics chosen.** Z-covtree, motivated by "asynchronous becoming" (p. 28), is "a consistent covariant framework for two-way growth"; in it "the event that the completed order contains a post is not measurable" (p. 43).

They weigh both sides (p. 42): the difficulties with alternating growth and the covariance result "may be considered as evidence against the existence of physically meaningful dynamical growth models for two-way infinite causal sets"; "On the positive side", non-deterministic models suggest "that there may be models in which the convex-events may yet form a rich and interesting class of observables." A later overview still asks whether growth should produce "only those which are past-finite" (Zalel 2023, p. 2).

A world-cycle's contraction is not a post, nor a partial one. The texts' contraction is bounded: fire burns below the Ābhassara heaven, water below Subhakiṇha, wind below Vehapphala, and "one buddha-field is always what is destroyed" (*sadāpi ekaṃ buddhakkhettaṃ vinassati*, AN-a on AN 4.156, s0402a.att.xml:5988), 10^{12} world-systems destroyed and re-formed together (*Visuddhimagga* XIII §404, e0102n.mul.xml:729–:733). Locality is therefore no contrast; Dowker and Zalel's partial posts and breaks are local too. The contrast is in who populates the new world: beings "fall from the Ābhassara body and come to this state" (*ābhassarakāyā cavitvā itthattaṃ āgacchanti*, DN 27, s0103m.mul.xml:1105), the Ābhassara lying outside the fire's reach, and the DN sub-commentary adds beings from "world-systems other than the contracting ones" (*saṃvaṭṭamānalokadhātūhi aññalokadhātūsu*, s0101t.tik.xml:1945). Read as a causal set, with each stream a chain and the destroyed field related to nothing outside it, the new world's past runs through moments outside the collapse. It is not closed, as a partial break requires, and those moments precede the new world without being related to the collapse, as a partial post forbids. **A contraction in the texts is neither a post nor a partial post or break.**

4.3 • Two axes together

THE BEGINNING: does the order of what has happened have a first member? (for a causal set, its past; in the texts, each stream of mind, no global one)		
WHAT EXISTS every born element is kept (growing block)	a first member	no first member
	Classical sequential growth (Rideout–Sorkin): grows from the empty set; order of birth = physical order TP: posts = bounces WITH a beginning Partial posts and breaks: local bounces (Dowker & Zalel)	Alternating growth (Bento, Dowker & Zalel): elements born to the PAST of existing ones; becoming given up "(nearly if not entirely)" ATP, the one covariant model: bounces with NO beginning; deterministic Models (18): not deterministic, not covariant Z-covtree: consistent, covariant; no dynamics chosen; a post not measurable
only the present (presentism about conditioned dhammas)	(a mind-stream with a first moment: the fortuitous-origination view, DN 1 §67–68, recorded to be rejected; Vism XVII §584: ignorance is not an uncaused root)	MAHĀVIHĀRA: presentism about conditioned dhammas; each mental state arises after its proximity condition; no stream of mind has a first member (SN 15 · Paṭṭhāna · AN 10.61); bodies and world-regions begin ← no causal-set model offered
Nibbāna and concepts are time-free (kālavimutta) and sit outside both axes.		

The Mahāvihāra's cell asks three things of a model: (1) each mental state arises after its proximity condition, and nothing past comes to be after the present, which is the order half of becoming in Sorkin's sense (the other half, that the born element "signifies that the event (of its birth) has already happened", p. 8, is the persistence §3 denies); (2) no stream of mind has a first member; (3) nothing past exists. No causal-set dynamics in the literature read holds all three, but (3) alone carries that verdict, since every causal set keeps its past (§4) and no one read here, this paper included (§6), builds a presentist one. The informative question is whether a model holds (1) and (2). Classical sequential growth holds (1) and not (2); alternating growth reaches (2) by giving up (1); on Z-covtree no dynamics has been chosen. On the beginning alone the question is open: a gap, not a finding against the physics.

The nearest counter to (1) is the *Paṭṭhāna*'s post-nascence condition, from a later mental state to earlier-arisen matter (the sibling paper, §3.3). That matter arose first and still exists when conditioned, so the order of arising is untouched.

5 • The Answer to Dowker

Dowker asked for "the Vibhajavadin view of time ... by examining the earliest existing records." The records give a three-part answer.

1. **There is no single Vibhajjavādin view of time, and the two referents converge on one school seen from two sides.** The name covered a family of schools (Cousins). In the northern sources it names a *present-plus-unripened-past* view, whose past half Vasumitra assigns to the Kāśyapiya, as, by Stcherbatsky's report, does Xuanzang's commentary for the view as a whole. The Pāli branch's own debate book refutes the past half of that view in its own words (§299), and its commentary names the holders *Kassapikā*. On the past, the northern and Pāli reports agree. On the future they differ: the *Kośa* denies it, and Vasumitra records only that no formation has a future cause, while the Pāli debate has the Kassapika grant existence to a future that is bound to arise.

2. **The view that resembles causal-set accretion most is the one the Pāli Vibhajjavādin record argues against, and the resemblance is partial.** Sorkin's instinct matched a real Buddhist position in keeping some past and denying the future. That position is the *Kośa's* Vibhajjavāda and the Kassapika's past half, not the Theravāda's, and its past thins where his is kept whole.
3. **The Theravāda's own view is presentism about conditioned dhammas**, grounded in the *Niruttipatha Sutta* and defended in the *Kathāvatthu*: whatever is past has ceased, and what has ceased does not exist. Nibbāna and concepts are time-free and fall outside the three times.

The records also bear on a question the footnote did not ask, whether the past has a beginning (§4.1–§4.3). This is a historical answer about what the records say. It takes no position on which view of time is true.

6 • The Contrast: Influence Without Existence

If the Theravāda is not a growing block, is there anything in it for a causal-set physicist? There is one thing, and it is a contrast rather than a correspondence.

A presentist owes an account of how the past can matter if it does not exist. The Pāli Abhidhamma gives one. Its analysis of conditioning is the *Paṭṭhāna*, summarized in *Abhidhammatthasaṅgaha* ch. 8. The *Abhidhammatthavibhāvinī-ṭīkā*, Sumaṅgala's sub-commentary on the *Saṅgaha* (c. 12th century, a later layer than the *Saṅgaha* itself), states the temporal reach of every one of the twenty-four conditions. In the CST text, its comment on ch. 8, §37 sorts them so:

TEMPORAL REACH (Vibhāvinī on Saṅgaha ch. 8 §37)	CONDITIONS	COUNT
past only (the moment just ceased)	proximity, contiguity, repetition, absence, disappearance	5
past or present (dvikālika)	kamma	1
all three times (tikālika)	object, predominance (object form), decisive support	3
present only (paccuppannakālika)	the other fifteen	15
		24
time-free (kālavimutta; Saṅgaha §37 root verse)	nibbāna, concepts (paññatti) – not a condition class	

- **Five conditions reach only the immediate past:** proximity, contiguity, repetition, absence and disappearance (*anantara, samanantara, āsevana, natthi, vigata*). They carry the moment that has just ceased into the next.
- **The kamma condition reaches past or present (dvikālika).** The *Saṅgaha's* own root text says that volition conditions its results across different moments (*nānākkhaṇikā cetanā*, ch. 8 §17): an action long gone conditions its result now. The past act does not exist when its result arises; the *Kathāvatthu* (§299) refutes the thesis that past action whose result has not ripened exists. **Nothing persists between them.** The commentary on §299 explains one of the opponent's assents by his doctrine (*laddhi*) that there exists a *kammūpacaya*, an accumulation of kamma "reckoned as the non-perishing of actions" (*kammānaṃ avināśasaṅkhāto*, [abh03a.att.xml:3225](#)), and the debate exists to refute the opponent's doctrine. The ceased act is a condition by kamma across moments, not a thing kept.
- **Three conditions reach all three times (tikālika):** the object condition, predominance (the *Vibhāvinī* names it without qualification; its object form, *ārammaṇādhīpati*, is the one that can reach past and future, since co-nascent predominance is present by definition) and decisive support. So the future also conditions, as an object of thought or intention.

- The other fifteen conditions are present-only.
- A time-free class stands outside the classification: nibbāna and concepts are *kālavimutta* (§3).

The *Kathāvatthu* adds the epistemological half of Dowker's question. At IX.6 (§559) an opponent holds that a consciousness whose object is past has no object; the commentary names the opponents the Uttarāpathakas and gives their reason, that since a past object does not exist, a mind that takes one must be objectless ([abh03a.att.xml:4629](#)). The Theravādin refutes this: a mind can turn toward the past, attend to it and aspire concerning it, so a past object is a genuine object even though it does not exist. **Perception and memory reach what is not.**

So the Theravāda's answer to "how does the past act?" is **influence without existence**: the past conditions the present, and is cognized by it, without being real now. Set beside a causal set, this is a **contrast**, and its use is that it locates the disagreement exactly. In causal sets the order relation, not a background time, carries the physics, and the order holds *between elements that remain in the set*; the past is kept. The Theravāda keeps the effects of conditioning and discards the past relatum; nothing persists to be related. The two agree that the past acts and disagree about whether it is there. A presentist causal set would be a different ontology from Sorkin's, and whether one is coherent is a question for physics, not for this paper.

7 · The Strongest Objection, Carried

Objection. "Presentism is a Western category, and the Abhidhamma is a phenomenology of experience. It is not a metaphysics of spacetime. Mapping *atthi* ('exists') onto the ontology of a causal set compares two different questions."

What survives it. Much of the objection stands, and the paper does not try to answer it away. The *Kathāvatthu*'s *atthi* is about dhammas, momentary mental and material phenomena, not about spacetime points. Its debates are about what may be *said* to exist in a doctrinal register. And the Abhidhamma's momentariness has no metric: its moments carry no length in any physical unit (the texts name the mind-moment as a unit of reckoning and time nothing in seconds; [abhidhamma-and-discrete-quantum-gravity](#) §5.2–§5.3), and its "present" is not a global simultaneity slice, which causal sets deny in any case. What survives is historical and exact: which positions on the existence of the past and the future were taken, by whom, and which the Pāli tradition rejected. Sorkin's footnote made a historical claim, and a historical claim can be checked against records. That, and placing the records against published models in §4.1–§4.3, is all this paper does with the physics.

The counter-argument to this paper's own strongest claim. Our strongest claim is that the Pāli Vibhajjavāda is Sorkin's foil. Against it, the provenance of Sorkin's word points away from the Pāli branch. Dowker's own pointer, Stcherbatsky p. 43, describes the *Kośa*'s school, and Stcherbatsky's footnote reports, after McGovern, Xuanzang's commentary identifying that school with the Kāśyapīyas. So the referent of "Vibhajavadin" in the causal-set literature is the northern one by construction, and Sorkin's footnote is broadly right about the school it names. On this reading, a comparison with the Mahāvihāra changes the subject. We accept this. We claim only that the branch that preserved the Pāli canon, and uses the name today, holds the opposite view; that its record refutes the named school's past half; and that the resemblance even for the named school is partial (§4). Whether a reader of the footnote should have the Mahāvihāra in mind is a question about the word, and the records show why it is ambiguous.

8 · What Would Falsify These Findings

- A primary Theravāda source that affirms the existence of unripened past kamma (as opposed to its conditioning force) would undo §3's reading of the Mahāvihāra as presentist.
 - A reading of Sorkin's sequential growth in which past elements cease to exist, rather than remain in the set, would move his view toward presentism and undo §4's placement. His own texts, read on the date of writing, say the opposite.
 - A reliable attribution of the §299 thesis to a school other than the Kassapikas, evidence that the Kośa's Vibhajyavādin denoted the Mahāvihāra specifically, or northern evidence separating Vasumitra's Kāśyapīya from the Kośa's Vibhajyavādin, would change §5.1.
 - A primary Theravāda source that treats past kamma, or a past object, as EXISTING when it conditions or is cognized (for example, one that accepts a *kammūpacaya*) would undo §6's "nothing persists".
 - A reading of *Kathāvatthu* IX.6 §559 on which the past object is a present surrogate, a memory-image existing now, would undo the epistemological half of §6.
 - The breaks stated under findings 6 and 7 would undo, respectively, §4.1's step and §4.2–§4.3's placement. A presentist causal-set dynamics would move the Mahāvihāra's cell in §4.3 on both axes.
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9 · Stake, and What Was and Was Not Done

The authors' stake. The authors write from within the Theravāda, and the institution publishing this paper grounds its values in the Pāli canon and has ruled Theravāda readings over Sarvāstivāda ones on its own questions. That creates a pull toward finding one's tradition anticipated by frontier physics. **This paper's first draft of the thesis made exactly that error.** It read the Pāli presentism as Sorkin's "becoming", on his side. An adversarial review, run before any prose was written, found that Sorkin's growth keeps the past, and the thesis was reversed. The reversal is the paper. A later review round corrected a second overreach of the same kind in the other direction: the first published text placed the Kośa's view at the growing block, which its own translation contradicts.

Done:

- The four time debates of the *Kathāvatthu*, the *Niruttipatha* quotations and §559, with their commentarial attributions, read in the Chaṭṭha Saṅgāyana text;
- the commentary's schism narrative and its gloss on §299 (*kammūpacaya*) read in the same text;
- the Kośa's definition read in Pradhan's Sanskrit;
- Stcherbatsky 1923, p. 43 and its footnote, read (archive.org, Digital Library of India copy);
- Vasumitra's *Samayabhedoparacanacakra*, the Kāśyapīya section, read in Xuanzang's Chinese (CBETA T49 no. 2031);
- Sorkin 2007 and Dowker 2022/2023 (both versions) read in full;
- the temporal classification of the conditions read in the CST text of the *Abhidhammatthavibhāvinī-ṭīkā* (a sub-commentary, not the *Saṅgaha* itself), and *nānākkhaṇika* and the time-free class checked in the *Saṅgaha* root;
- the census run as described;
- for the revision of 2026-09-30: every CST passage cited in §4.1–§4.2 (citations 24–30, 33), read in its surrounding lines;

- Bento, Dowker and Zalel read in full (arXiv v1, 22 September 2021); Dowker and Zalel (2017) pp. 1–6 and (2023) pp. 1–4; Martín et al. (2001) pp. 1–3; Zalel (2023) pp. 1–2;
- `kala/Time.lean` built with Lean 4.34.1 on 2026-09-30: seven theorems and the model check, and all six deliberate breaks fail as required;
- for the revision of 2026-10-06: the four time debates, their commentary, §559 and its commentary, the Third Council passage and the schism list re-read in the CST text; the *Kośa* passage re-read in GRETEL; Vasumitra's Kāśyapīya section re-read in CBETA; Sorkin 2007, Dowker (both versions), Bento, Dowker and Zalel, Stcherbatsky 1923, Cousins 2001 and Arageorgis 2016 re-opened; Broad 1923 pp. 66–67 read (listed under *Sources checked*).

Not done:

- The Sarvāstivāda *Vibhāṣā* and the Chinese Āgama parallels (where the Kāśyapīya view is also reported) were not consulted.
- *Points of Controversy*, Pruden's translation of the *Kośa*, Cox's *Disputed Dharmas* (1995) and Bureau's *Les sectes bouddhiques du Petit Véhicule* (1955) were not read; they are named as the standard places to check this paper, not as sources for it.
- Cousins was read in English; the Mahāvihāra's lineage texts (the *Dīpavaṃsa*) were read only as quoted by the commentary.
- Later Theravāda exposition of kamma (for example, modern Burmese teaching of a kammic potency in the mental continuum) was not read (§10).
- The physics of a presentist causal set was not attempted.
- For the revision: Rideout and Sorkin (2000) was not re-read; its content is reported as Bento, Dowker and Zalel state it. Their journal version was not compared with arXiv v1, and the thesis they cite for random forward and backward growth (their ref. [28]) was not read. The search for later past-infinite dynamics was of the English web and arXiv only (2026-09-30). Later Theravāda cosmology and the Sarvāstivāda account of the aeons were not consulted.

10 · Honest Limits

- **The *Kathāvatthu*'s date is traditional.** It is ascribed to the Third Council (3rd century BCE), and its irregular structure has been read as growth over time, new controversies added as they arose (von Hinüber 2000, p. 71, §145, as quoted in a secondary source; not read directly for this paper). The four time debates are in its first chapter. The commentary that supplies the self-designation and the opponents' names is c. 5th century CE. "Earliest existing records" here means the earliest the Pāli tradition preserves, not the earliest that existed.
- **The attributions are the commentary's.** The *Kathāvatthu* names no opponents. A later author's attribution is evidence about the tradition's self-understanding, not a contemporaneous report. The same commentary's schism list files the Kassapikas under the Sabbatthivāda (§2.2).
- **The *Kośa*'s definition is a Sarvāstivādin doctrinal report.** Vasubandhu reports the Vibhajyavādins as the Sarvāstivāda's opponents in a polemical context. Vasumitra's agreement on its past half and the *Kathāvatthu*'s own wording at §299 corroborate its content, but no text of the Kāśyapīyas themselves is quoted here.
- **"Exists" may be an efficacy predicate.** In the three-times controversy, existence is often keyed to causal efficacy: Stcherbatsky glosses the *Kośa* school's existent past as the part that "had not already lost [its] influence", and Miller reports the Sarvāstivāda marking the present as the causally efficacious moment. Where *exists* means *efficacious*, the *Kośa*'s existent unripened karma and the Theravāda's non-existent but conditioning karma may

differ more in wording than in ontology. §6's contrast holds only if the *Kathāvatthu's atthi* is not an efficacy predicate; this paper reads it as not one, because the *Kathāvatthu* denies existence to exactly what it grants as a condition, but that reading is an interpretation.

- **"Nothing persists" is stated at the layer of the *Kathāvatthu* and its commentary.** Later Theravāda exposition speaks of a kammic potency carried in the mental continuum. Whether that potency is a persisting store of the kind the commentary refuses, or a way of speaking about conditioning, was not examined.
- **The time-free class.** "Presentism" in this paper is presentism about conditioned dhammas. Nibbāna and concepts are *kālavimutta* and fall outside the comparison. A reader who treats the Theravāda as presentist without that qualification misstates it.
- **The physics mapping is a mapping of positions on existence and, since the revision, on the beginning.** It does not claim that any Buddhist school anticipated a theory of quantum gravity, and it makes no claim about which theory of time is correct.
- **One edition.** Paragraph numbers are the Chaṭṭha Saṅgāyana's; in the first chapter its numbering of the debates runs one lower than the numbering in common use (§3). A reader should locate passages by § number and title.
- **The census was quick and English-only.** The field of Buddhism-and-physics comparison is large and poorly indexed. Absence from our search is weak evidence, and the census's own second NOT FOUND fell to a book it could not read (Stcherbatsky). Findings 6 and 7 were not searched at all.
- **The beginning claim rests on a reading of rows.** Its premise is read off the *Paṭṭhāna's* general statement of the proximity condition and its negative method; the commentaries read it so too (§4.1), but no proof assistant can check a reading. The ordinal model of a stream is ours. `kala/Time.lean` is exploratory and outside the repository's pre-registration, and the two runs that record the reasoning (`runs/2026-09-30-the-beginning` , `runs/2026-09-30-longest-chain`) are retrospective and count for nothing as tests.
- **The predicate line is an interpretation.** That a beginningless succession of impermanent states affirms neither the permanence nor the annihilation MN 63's questions ask about is this paper's reading, not a text's.
- **Streams as chains is a comparison.** §4.2's infinite pasts, and its unclosed past of a re-evolved world, hold only if a stream's moments are read as a chain in a causal set (the sibling paper's relation *S*), the second also only if the destroyed field is related to nothing outside it.
- **"Growing block" is a contested label for the physics.** Earman (2008) reads causal-set growth as a growing block; Arageorgis (2016) argues that discrete general covariance leaves it no better placed than continuum relativity. The comparison in §4 rests on one feature of the models, that a born element is never removed, and not on the label; Earman was read only as Arageorgis quotes him.
- **The physics placement is dated.** It rests on one paper read in full (Bento, Dowker and Zalel) and four read in part. It is not a no-go theorem: its authors weigh their negative results as evidence, not proof, and their positive ones as encouragement.

11 • Close

Sorkin heard, in a school name, an echo of the time his models grow. The echo is real and partial. It comes from one school of a family, the one the northern sources describe, whose past is kept only while its fruit is owed. The branch that kept the Pāli canon answered that view in its canonical book of debates, with a question it asks of every past thing: *has it not ceased?* The answer to Dowker's invitation is that the earliest records disagree with each other on the existence of the past — the question on which, in a different register, the physics of becoming is still open. On a second question the Pāli records read here agree: no stream of mind has a first member, though bodies and world-

regions begin. There the physics, in the one study read closely, has a consistent covariant frame for a past without beginning and no dynamics on it yet chosen.

Cross-Venue References

Venue	Identifier
Primary canonical	https://thonly.org/research/the-vibhajjavadin-view-of-time
GitHub	https://github.com/thonly/publications/blob/main/defensive-publications/the-vibhajjavadin-view-of-time.md
Zenodo (concept DOI, resolving to the latest version)	https://doi.org/10.5281/zenodo.23020015
Internet Archive (the site, captured daily)	https://web.archive.org/web/2026*/thonly.org/research/the-vibhajjavadin-view-of-time
Software Heritage (the repository)	https://archive.softwareheritage.org/browse/origin/?origin_url=https://github.com/thonly/publications
Census pre-registration (public)	<code>thonly/publications</code> commit <code>7348ba2</code> , <code>timestamps/census/physics-lens-prereg.md</code>
Sibling paper (<code>abhidhamma-and-discrete-quantum-gravity</code>)	https://thonly.org/research/abhidhamma-and-discrete-quantum-gravity
Lean check of §4.1 (public, CC0)	https://github.com/SiliconWat/formal-abhidhamma — <code>kala/Time.lean</code> , theorem <code>beginningless</code> ; software concept DOI 10.5281/zenodo.23067241
Independent timestamps	an OpenTimestamps proof anchored in Bitcoin, and RFC 3161 tokens from three timestamp authorities, one of them eIDAS-qualified; a timestamp proves that this exact text existed by its date and nothing about authorship, originality, or validity

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- Earman (2008) and Tooley (1997): Crossref records only; their content is cited as Arageorgis reports it.
- Broad (1923): <https://archive.org/details/scientificthough00broauoft> (the 1923 printing, New York and London; pp. 66–67).
- Stcherbatsky (1923): <https://archive.org/details/in.ernet.dli.2015.99616> (p. 43 and its footnote).
- Cousins (2001): the article's PDF, linked from https://buddhistuniversity.net/content/articles/on-vibhajjavadins_cousins (the summary, the two senses known to the *ṭikā* writers, the common ancestors of four schools, and the double meaning of the Third Council story); DOI from Crossref.
- Miller (2017): Crossref record; the sentences reported in the Prior-Art Statement were checked against her preprint at the review of 2026-09-28 and not re-opened.
- Gauvain (n.d.): checked at the review of 2026-09-28; not re-opened.

- Vasumitra, T49 no. 2031: the CBETA XML (<https://github.com/cbeta-org/xml-p5>, T/T49/T49n2031.xml), the Kāśyapīya section in full.
- *Abhidharmakośabhāṣya*: the GRETIL plain-text e-text, Pradhan pp. 296–297 (the two definitions, the four kinds of Sarvāstivādin, and the verdict on the third).
- von Hinüber (2000): the page, paragraph and the sentence on the *Kathāvatthu*'s growth as quoted in the English Wikipedia article "Kathāvatthu" (a secondary source); the 1996 first edition from Crossref. The book was not opened.
- The numbering of the *Kathāvatthu*'s debates in common use: SuttaCentral's index (debates I.6–I.10 and IX.6).
- The blog post of 2026-07-31: opened; it describes itself as AI-generated content. ResearchGate publication 397427972 refused automated access on 2026-10-06; it is cited as the census of 2026-09-27 found it.
- *Points of Controversy*, Pruden's translation, Stcherbatsky (1932, cited as Dowker gives it), Ricard & Thuan and the Dalai Lama are cited as before and were not re-opened.
- The Pāli passages were read in the Chaṭṭha Saṅgāyana edition (VRI tipitaka-xml, romn/, commit 05d5d3c) through a local reader of that text, each with the lines around it: *Kathāvatthu* §75, §282–§283, §296–§300, §304–§305 and §559–§560; its commentary on §282, §297–§299 (with the *kammūpacaya* gloss, abh03a.att.xml:3225), §304 and §559–§561, its Third Council account (:1949 , :2181) and its schism list with the *Dīpavaṃsa* verses it quotes (:1961); the *Abhidhammatthasaṅgaha*, ch. 8 §17 and the §37 verse, and the *Vibhāvīnī* on §37 (abh07t.nrf.xml:6793); and every passage cited in §4.1–§4.2 (citations 24–30 and 33), with the *Visuddhimagga*'s AN 10.61 citation located at e0102n.mul.xml:3557 .